



NAME: \_\_\_\_\_

TEACHER: \_\_\_\_\_

## ***GOSFORD HIGH SCHOOL***

**2015**

### **EXTENSION 2 MATHEMATICS HSC ASSESSMENT TASK 2.**

**Time Allowed:** 90 minutes (plus 5 min. reading time)

- Write using black or blue pen.
- Board-approved calculators may be used.
- Questions 7, 8, 9 and 10 should all be started on a new page.
- All necessary working should be shown in Questions 7, 8, 9 and 10.

| QUESTION | QUESTION TYPE     | MARKS | RESULT |
|----------|-------------------|-------|--------|
| 1-6      | MULTIPLE CHOICE   | 6     |        |
| 7        | EXTENDED RESPONSE | 15    |        |
| 8        | EXTENDED RESPONSE | 12    |        |
| 9        | EXTENDED RESPONSE | 15    |        |
| 10       | EXTENDED RESPONSE | 12    |        |
|          | TOTAL             | 60    |        |

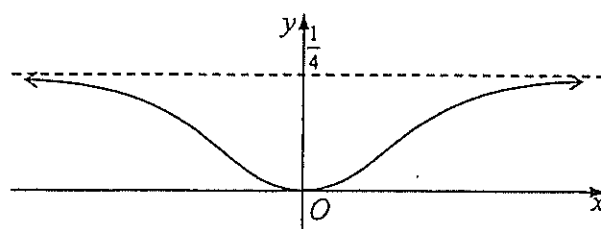
MULTIPLE CHOICE (6 marks). Answer on the multiple choice answer sheet.

1. Let  $z = 5 - i$  and  $w = 2 + 3i$ .

What is the value of  $2z + \bar{w}$ ?

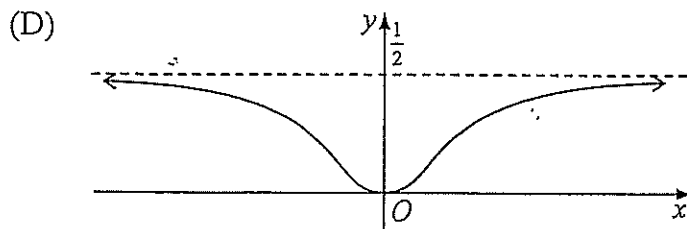
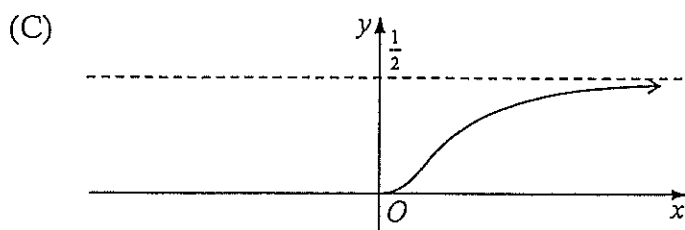
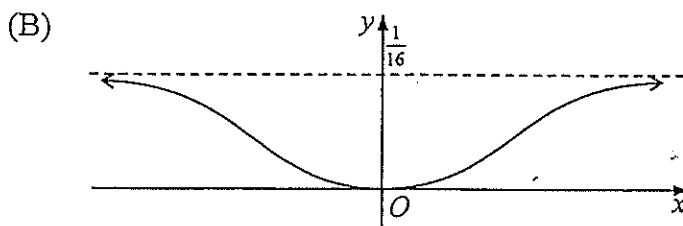
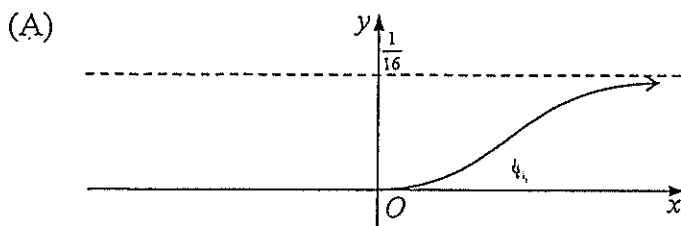
- (A)  $12 + i$ .      (B)  $12 + 2i$ .      (C)  $12 - 4i$ .      (D)  $12 - 5i$ .

2. The diagram shows the graph of  $y = f(x)$ .



DIAGRAMS NOT  
TO SCALE.

Which of the following best represents the graph of  $y = \sqrt{f(x)}$ ?



3. The equation  $2x^3 - 3x^2 - 5x - 1 = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

What is the value of  $\frac{1}{\alpha^3\beta^3\gamma^3}$ ?

- (A)  $\frac{1}{8}$ . (B)  $\frac{-1}{8}$ . (C) 8. (D) -8.

4. What is the eccentricity of the ellipse  $4x^2 + 6y^2 = 24$ ?

- (A)  $\frac{\sqrt{10}}{2}$ . (B)  $\frac{\sqrt{15}}{3}$ . (C)  $\frac{\sqrt{3}}{3}$ . (D)  $\frac{\sqrt{13}}{3}$ .

5. The cube roots of unity are 1,  $\omega$  and  $\omega^2$ . Simplify  $(1 - \omega + \omega^2)(1 + \omega - \omega^2)$ .

- (A) 0. (B) 1. (C) 2. (D) 4.

6. If  $\frac{4x}{x^2-x-12} \equiv \frac{a}{x-4} + \frac{b}{x+3}$ , then

- (A)  $a = 16, b = 12$ . (B)  $a = 12, b = 16$ . (C)  $a = \frac{16}{7}, b = \frac{12}{7}$ . (D)  $a = \frac{12}{7}, b = \frac{16}{7}$ .

**Question 7.** (15 marks) Start a new page.

(a) If  $z = 2 + i$  and  $\omega = 1 - 3i$  find in the form  $x + iy$

(i)  $z^2$ . (1)

(ii)  $z\bar{\omega}$ . (1)

(iii)  $\frac{z}{\omega}$ . (1)

(b)

(i) Express  $z = 1 + \sqrt{3}i$  in modulus-argument form. (2)

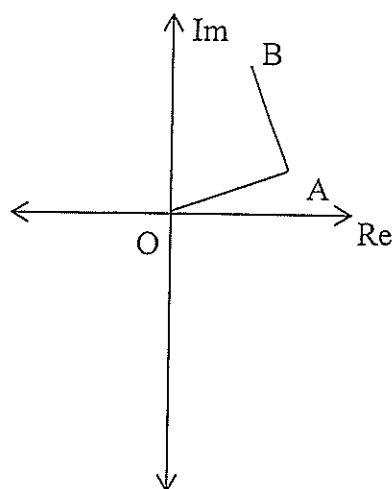
(ii) Show that  $(1 + \sqrt{3}i)^6$  is a real number. (2)

(c) For the complex number  $z = x + iy$ , where  $x$  and  $y$  are real numbers, find and clearly sketch the curve on an Argand diagram for which

(i)  $|z + \bar{z}| \leq 2$ . (2)

(ii)  $|z - i| = \sqrt{2}|z + i|$  (3)

(d) The point A in the Argand diagram below represents the complex number  $z = a + ib$ . The point B represents the complex number  $2 + 5i$ .

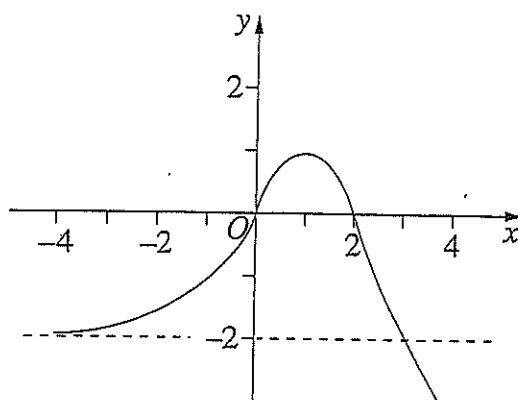


NOT TO SCALE

If the complex number represented by the point C is such that OABC is a square, find C in terms of  $a$  and  $b$  and hence evaluate  $a$  and  $b$ . (3)

**Question 8** (12 marks) Start a new page.

(a) The graph of  $y = f(x)$  is shown below.



Draw a neat sketch of each of the following on the template sheet provided.

(i)  $y = |f(x)|.$  (1)

(ii)  $y = [f(x)]^2.$  (2)

(iii)  $y = f(|x|).$  (1)

(iv)  $y^2 = f(x).$  (2)

(v)  $y = \frac{1}{f(x)}.$  (3)

(b) The equation of a curve is  $4x^2 + xy + y^2 = 10$ . Find the equation of the tangent to the curve at the point (1,2) on it. (3)

**MAKE SURE THAT YOU ATTACH QUESTION 8 (b) TO THE BACK OF THE TEMPLATE SHEET PROVIDED FOR QUESTION 8(a).**

**Question 9** (15 marks) Start a new page.

(a) If  $\alpha$ ,  $\beta$  and  $\gamma$  are the roots of the equation  $x^3 - 7x^2 - 7 = 0$  find the polynomial equation whose roots are  $\alpha^2$ ,  $\beta^2$ ,  $\gamma^2$ . (3)

(b) Express  $\frac{2}{x^3+2x}$  in the form  $\frac{A}{x} + \frac{Bx+C}{x^2+2}$ . (2)

(c) Consider the equation  $z^4 + pz^3 + qz + r = 0$ , where  $p, q$  &  $r$  are real numbers. The sum of the roots of this equation is 6 more than the product of the roots. If  $1 + i$  is a root of the equation, find  $p, q$  &  $r$ . (3)

(d) (i) Use DeMoivre's Theorem to express  $\cos 4\theta$  and  $\sin 4\theta$  in powers of  $\cos \theta$  and  $\sin \theta$ . Hence express  $\tan 4\theta$  as a rational function of  $t$ , where  $t = \tan \theta$ . (4)

(ii) Hence solve the equation  $t^4 + 4t^3 - 6t^2 - 4t + 1 = 0$ . (3)

**Question 10** (12 marks) Start a new page.

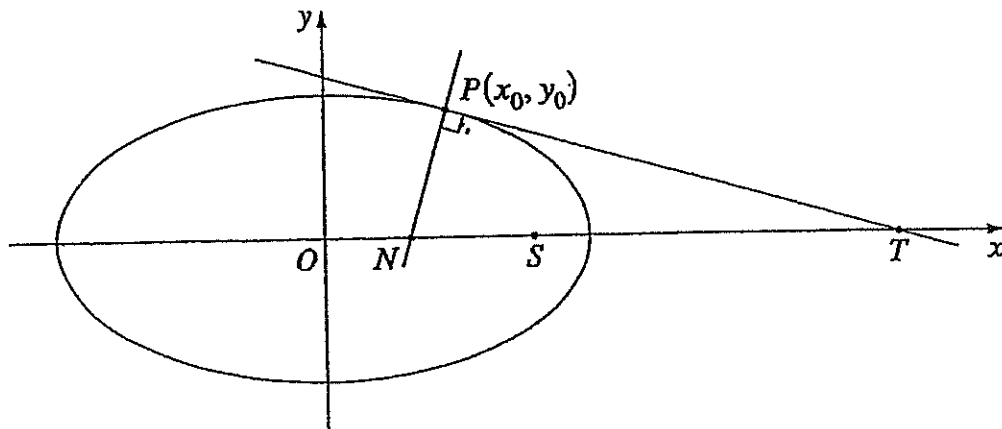
(a) Consider the ellipse  $\mathcal{E}$ , with equation  $\frac{x^2}{100} + \frac{y^2}{64} = 1$ .

(i) Calculate the eccentricity of  $\mathcal{E}$ . (1)

(ii) Find the coordinates of the foci and the equations of the directrices of  $\mathcal{E}$ . (2)

(iii) Draw a neat sketch of  $\mathcal{E}$  showing all important features. (2)

(b) The diagram shows the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ , with  $a > b$ . The ellipse has focus  $S$  and eccentricity  $e$ . The tangent to the ellipse at  $P(x_0, y_0)$  meets the  $x$ -axis at  $T$ . The normal at  $P$  meets the  $x$ -axis at  $N$ .



(i) Show that the tangent to the ellipse at  $P$  is given by the equation

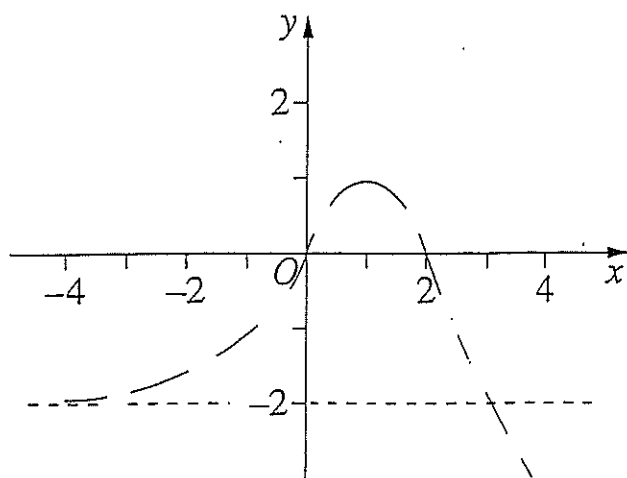
$$y - y_0 = - \frac{b^2 x_0}{a^2 y_0} (x - x_0). \quad (2)$$

(ii) Show that the  $x$ -coordinate of  $N$  is  $x_0 e^2$ . (2)

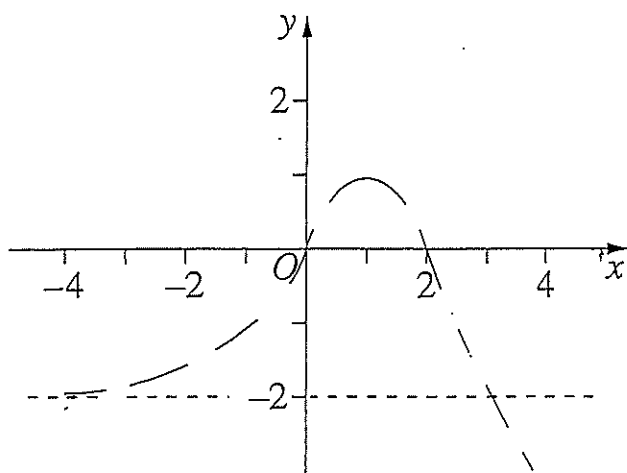
(iii) Show that  $ON \times OT = OS^2$ . (3)

Question 8(a) TEMPLATE SHEET

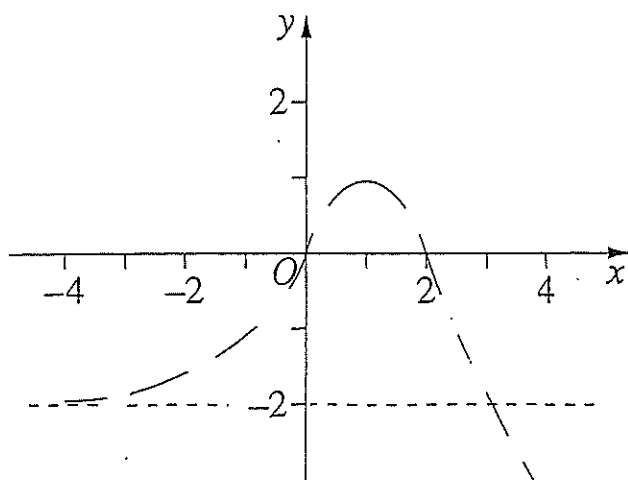
(i)



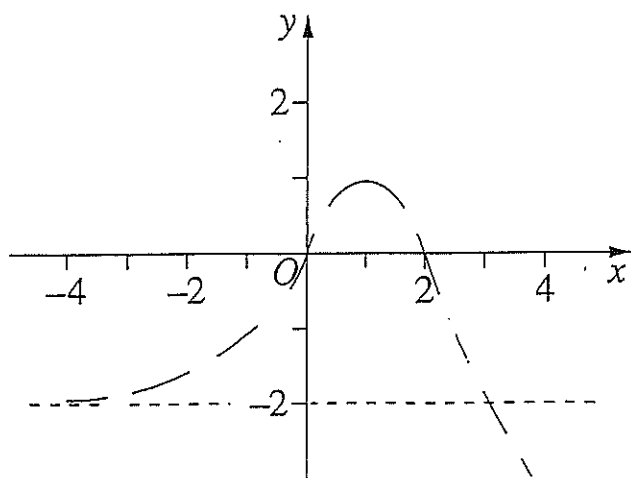
(ii)



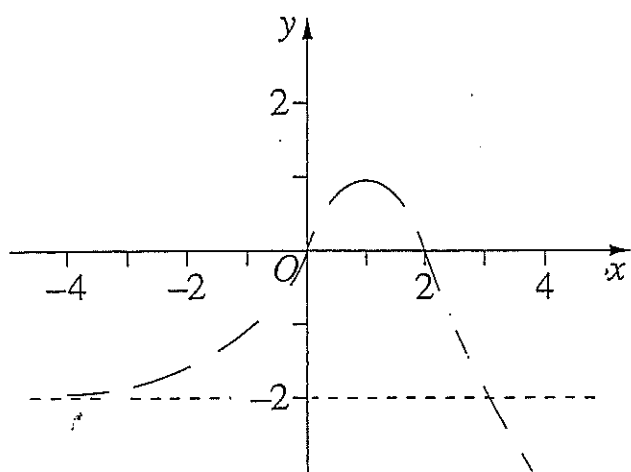
(iii)



(iv)



(v)



# MULTIPLE CHOICE

11.  $z + \bar{z} = 10 - 2i + 2 - 3i$   
 $= 12 - 5i$

Hence

(D)

12. (D) NB  $\sqrt{1/4} = 1/2$

13.  $\angle \beta \gamma = -\frac{\alpha}{a}$   
 $= -\frac{-1}{2}$   
 $= \frac{1}{2}$

$\alpha^3 \beta^3 \gamma^3 = \frac{1}{8}$

$\frac{1}{\alpha^3 \beta^3 \gamma^3} = 8$

Hence

(C)

14.  $4x^2 + 6y^2 = 24$   
 $\frac{x^2}{6} + \frac{y^2}{4} = 1$

$b^2 = a^2(1 - e^2)$

$4 = 6(1 - e^2)$

$\frac{2}{3} = 1 - e^2$

$e^2 = 1 - \frac{2}{3}$

$e^2 = \frac{1}{3}$

$e = \frac{1}{\sqrt{3}}$

$e = \frac{1}{\sqrt{3}}$

Hence

(C)

15. If  $z^3 - 1 = 0$

$\sum \text{roots} = -\frac{b}{a}$   
 $= 0$

$\therefore 1 + \omega + \omega^2 = 0$

$1 + \omega^2 = -\omega$

$\times 1 + \omega = -\omega^2$

$\therefore (1 - \omega + \omega^2)(1 + \omega - \omega^2)$

$= -2\omega \times -2\omega^2$

$= 4\omega^3$

$= 4$

Hence (D)

16. If  $\frac{4x}{x^2 - x - 12} = \frac{a}{x-4} + \frac{b}{x+3}$

$4x = a(x+3) + b(x-4)$

If  $x = -3$

$-12 = 0 - 7b$

$7b = 12$

$b = \frac{12}{7}$

If  $x = 4$

$16 = 7a$

$a = \frac{16}{7}$

Hence

(C)

# Question 7

a)  $z = 2 + i$ ,  $w = 1 - 3i$

(i)  $z^2 = (2 + i)^2$   
 $= 4 + 4i + i^2$   
 $= 3 + 4i$

(1)

(ii)  $z\bar{w} = (2 + i)(1 + 3i)$   
 $= 2 + 6i + i + 3i^2$   
 $= -1 + 7i$

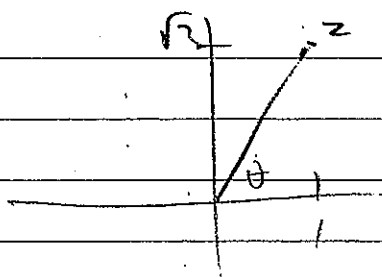
(1)

(iii)  $\frac{z}{w} = \frac{(2 + i)}{1 - 3i} \times \frac{(1 + 3i)}{1 + 3i}$   
 $= \frac{2 + 6i + i + 3i^2}{1 - 9i^2}$   
 $= \frac{-1 + 7i}{10}$

(1)

$= -\frac{1}{10} + \frac{7}{10}i$

b) (i)



$|z| = \sqrt{(\sqrt{3})^2 + 1^2}$   
 $= 2$

$\arg(z) = \tan^{-1} \sqrt{3}$   
 $= \frac{\pi}{3}$

$\therefore 1 + \sqrt{3}i = 2 \cos \frac{\pi}{3}$

(2)

(ii)  $(1 + \sqrt{3}i)^6 = \left(2 \cos \frac{\pi}{3}\right)^6$

$= 2^6 \cos 2\pi$

$= 2^6 \times \cos 0$

$= 64 + (\cos 0 + i \sin 0)$

$= 64 + 1$

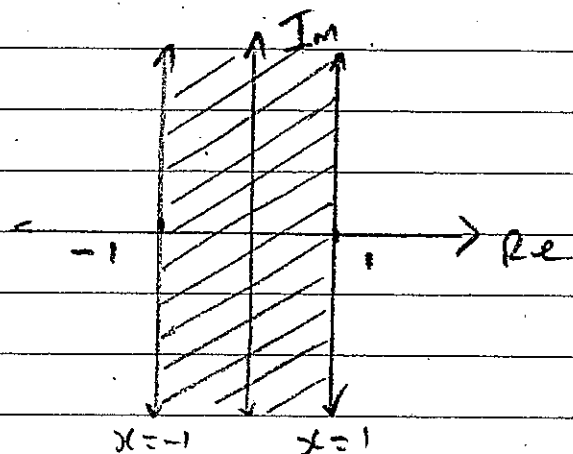
$= 64$  which is real.

c) (i)  $z = x + iy$ ,  $\bar{z} = x - iy$

$$\therefore z + \bar{z} = 2x$$

$$\text{So } |2x| \leq 2$$

$$|x| \leq 1$$



(2)

(ii) Let  $z = x + iy$

$$|2 - i| = \sqrt{2} |z + i|$$

$$|x + i(y-1)| = \sqrt{2} |x + i(y+1)|$$

$$\sqrt{x^2 + (y-1)^2} = \sqrt{2} \sqrt{x^2 + (y+1)^2}$$

$$\therefore x^2 + (y-1)^2 = 2(x^2 + (y+1)^2)$$

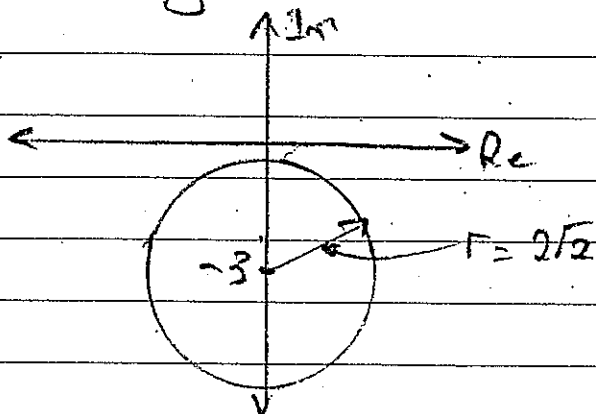
$$x^2 + y^2 - 2y + 1 = 2(x^2 + y^2 + 2y + 1)$$

$$x^2 + y^2 - 2y + 1 = 2x^2 + 2y^2 + 4y + 2$$

$$0 = x^2 + y^2 + 6y + 1$$

$$x^2 + y^2 + 6y + 9 = 8$$

$$x^2 + (y+3)^2 = 8$$



(3)

c) If A is  $z = a+ib$   
 C is  $\bar{z} = a\bar{i} + i\bar{b}$   
 $= -b+ia$ . (1)

Now  $2+5i = a+ib + -b+ia$   
 $= a-b + i(a+b)$

$$a-b=2$$

$$a+b=5$$

$$2a=7$$

$$a = \frac{7}{2}$$

(2)

$$b = \frac{3}{2}$$

Question 8a) See Template attached.

Q 8 b) If  $4x^2 + xy + y^2 = 10$

$$8x + y + x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$$

$$8x + y + \frac{dy}{dx} (x+2y) = 0$$

$$\frac{dy}{dx} = -\frac{(8x+y)}{x+2y}$$

If  $x=1, y=2$

$$m = -\frac{(8+2)}{1+4}$$

$$= -2$$

$\therefore$  eqn is  $y-2 = -2(x-1)$

$$y-2 = -2x+2$$

$$2x+y-4=0$$

(3)

## Question 9.

a) Let  $y = x^2$

since  $x = \alpha, \beta, \gamma$

$$y = \alpha^2, \beta^2, \gamma^2$$

$$\therefore x = \sqrt{y}$$

$$(\sqrt{y})^3 - 7(\sqrt{y})^2 - 7 = 0$$

$$y^{3/2} - 7y - 7 = 0$$

$$y^{3/2} = 7y + 7$$

$$y^3 = (7y + 7)^2$$

$$y^3 = 49y^2 + 98y + 49$$

ie  $y^3 - 49y^2 - 98y - 49 = 0$

(3)

$\therefore$  Required eq<sup>n</sup> is

$$x^3 - 49x^2 - 98x - 49 = 0.$$

b) If  $\frac{2}{x(x^2+2)} = \frac{A}{x} + \frac{Bx+C}{x^2+2}$

$$2 = A(x^2+2) + x(Bx+C)$$

$$2 = Ax^2 + 2A + Bx^2 + Cx$$

Let  $x=0$

$$2 = 2A \Rightarrow A = 1$$

Let  $x=1$

$$2 = A + 2A + B + C$$

$$2 = 1 + 2 + B + C$$

$$B + C = -1$$

(1)

$$\therefore -10 + q - 3p = 0$$

$$q - 3p = 10 \quad \text{--- (1)}$$

$$q + 2p = 0 \quad \text{--- (2)}$$

$$\textcircled{2} - \textcircled{1}$$

$$5p = -10$$

$$p = -2$$

$$\therefore q = 4$$

$$r = -4$$

$$\text{Soln is } p = -2, q = 4, r = -4.$$

$$d) \text{ (i) } (\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta.$$

$$\text{Also } (\cos \theta + i \sin \theta)^4 = \cos^4 \theta + 4 \cos^3 \theta i \sin \theta + 6 \cos^2 \theta i^2 \sin^2 \theta + 4 \cos \theta i^3 \sin^3 \theta + i^4 \sin^4 \theta$$

$$\text{RHS} = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta + i (4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta)$$

$$\text{Hence } \cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta \quad \checkmark$$

$$\sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta \quad \checkmark$$

$$\tan 4\theta = \frac{4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}{\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta}$$

$$\div \text{divided by } \cos^4 \theta$$

$$\begin{aligned} \tan 4\theta &= \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta} \\ &= \frac{4t - 4t^3}{1 - 6t^2 + t^4} \end{aligned}$$

4

ii) If  $\tan 4\theta = 1$

$$1 = \frac{4t - 4t^3}{1 - 6t^2 + t^4}$$

$$1 - 6t^2 + t^4 = 4t - 4t^3$$

ie  $t^4 + 4t^3 - 6t^2 - 4t + 1 = 0$

3

When  $\tan 4\theta = 1$

$$4\theta = \frac{\pi}{4}, \quad \pi + \frac{\pi}{4}, \quad 2\pi + \frac{\pi}{4}, \quad 3\pi + \frac{\pi}{4}$$

$$= \frac{\pi}{4}, \quad \frac{5\pi}{4}, \quad \frac{9\pi}{4}, \quad \frac{13\pi}{4}, \quad \dots$$

$$\theta = \frac{\pi}{16}, \quad \frac{5\pi}{16}, \quad \frac{9\pi}{16}, \quad \frac{13\pi}{16}$$

$$\therefore t = \tan \frac{\pi}{16}, \quad \tan \frac{5\pi}{16}, \quad \tan \frac{9\pi}{16}, \quad \tan \frac{13\pi}{16}$$

$$\text{or } t = \tan \frac{\pi}{16}, \quad \tan \frac{5\pi}{16}, \quad \text{or } \tan \frac{-7\pi}{16}, \quad \tan \frac{-3\pi}{16}$$

Question 10.

a) (i)  $b^2 = a^2(1 - e^2)$

$$64 = 100(1 - e^2)$$

$$\frac{64}{100} = 1 - e^2$$

$$e^2 = \frac{36}{100}$$

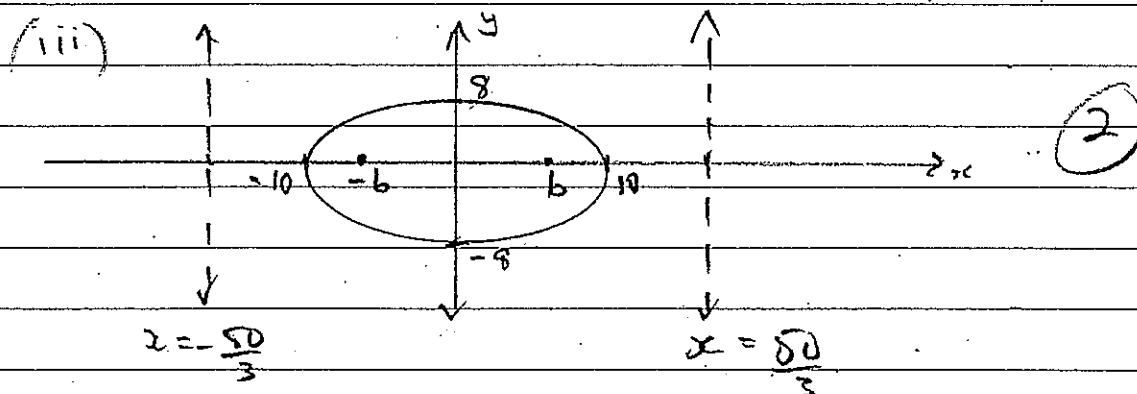
$$e = \frac{3}{5}, e > 0$$

(ii) Foci are  $(\pm ae, 0)$

ie  $(\pm b, 0)$

Directrices are  $x = \pm \frac{a}{e}$

ie  $x = \pm \frac{50}{3}$



b) (i) If  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

by implicit differentiation

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{2y}{b^2} \cdot \frac{dy}{dx} = -\frac{2x}{a^2}$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \times \frac{b^2}{2y}$$

$$\text{Let } x = -1, \quad 2 = A + 2A + B - C$$

$$2 = 1 + 2 + B - C$$

$$B - C = -1 \quad (2)$$

$$(1) + (2)$$

$$2B = -2$$

$$B = -1$$

$$C = 0$$

2

$$\therefore \frac{2}{x^3 + 2x} = \frac{1}{x} - \frac{x}{x^2 + 2}$$

c) Since  $1+i$  is a root we sub  $1+i$  into the eqn

$$\text{N.B. } (1+i)^2 = 1 + 2i + i^2$$

$$= 2i$$

$$(1+i)^3 = 2i(1+i)$$

$$= -2 + 2i$$

$$(1+i)^4 = (2i)^2$$

$$= -4$$

$$\text{Now } (1+i)^4 + p(1+i)^3 + q(1+i) + r = 0$$

$$-4 + p(-2+2i) + q(1+i) + r = 0$$

$$\text{Now } \sum \alpha = -p = \sum \alpha \beta \gamma \delta = +r$$

$$\text{So } -p = 6 + r$$

$$r = -p - 6$$

$$\therefore -4 + p(-2+2i) + q(1+i) - p - 6 = 0$$

$$-10 - 2p + 2pi + q + qi - p - 6 = 0$$

$$-16 + q - 3p + i(2p+q) = 0$$

At  $(x_0, y_0)$  the grad. of the tangent is

$$y' = \frac{-2x_0}{a^2} + \frac{b^2}{2y_0}$$
$$= -\frac{b^2 x_0}{a^2 y_0}$$

(2)

$\therefore$  the eqn of the tangent is

$$y - y_0 = -\frac{b^2 x_0}{a^2 y_0} (x - x_0)$$

(ii) the grad. of the normal is  $\frac{a^2 y_0}{b^2 x_0}$

$\therefore$  the eqn of the normal is

$$y - y_0 = \frac{a^2 y_0}{b^2 x_0} (x - x_0)$$

At N,  $y = 0$

$$\therefore -y_0 = \frac{a^2 y_0}{b^2 x_0} (x - x_0)$$

$$\frac{a^2 y_0}{b^2 x_0} x = \frac{a^2 y_0}{b^2} - y_0$$

$$\frac{a^2 y_0}{b^2 x_0} x = y_0 \left( \frac{a^2}{b^2} - 1 \right)$$

$$x = y_0 \left( \frac{a^2}{b^2} - 1 \right) \times \frac{b^2 x_0}{a^2 y_0}$$

$$x = x_0 \left( 1 - \frac{b^2}{a^2} \right)$$

Since  $b^2 = a^2 (1 - e^2)$

$$b^2 = a^2 - a^2 e^2$$

$$a^2 e^2 = a^2 - b^2$$

$$e^2 = 1 - \frac{b^2}{a^2}$$

Hence  $x = e^2 x_0$

2

(iii) From (i), when  $y=0$

$$0 - y_0 = -\frac{b^2 x_0}{a^2 y_0} (x - x_0)$$

$$a^2 y_0^2 = b^2 x x_0 - b^2 x_0^2 \quad *$$

$$\text{Since } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1$$

$$b^2 x_0^2 + a^2 y_0^2 = a^2 b^2$$

$$\therefore a^2 b^2 = b^2 x x_0 \quad \text{from } *$$

$$\frac{a^2 b^2}{b^2 x_0} = x$$

$$x = \frac{a^2}{x_0}$$

$$T \text{ is } \left( \frac{a^2}{x_0}, 0 \right)$$

$$\text{Now } ON \times OT = e^2 x_0 \times \frac{a^2}{x_0}$$

$$= a^2 e^2$$

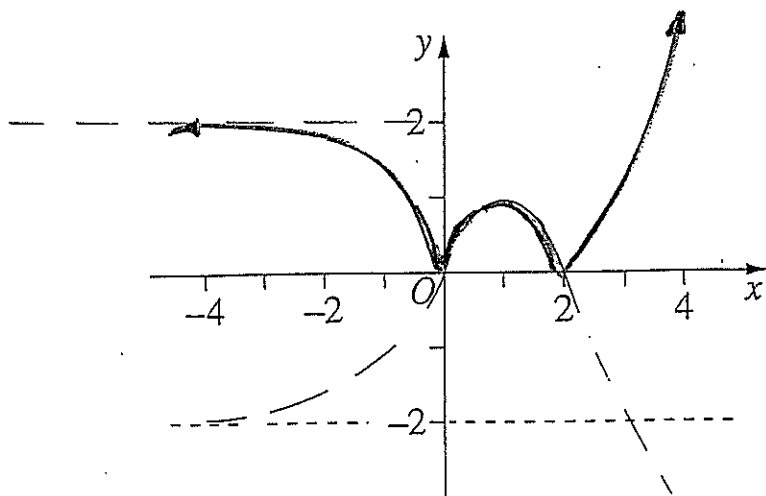
$$\therefore OS^2 = (ae)^2 = a^2 e^2$$

$$\therefore ON \times OT = OS^2$$

3

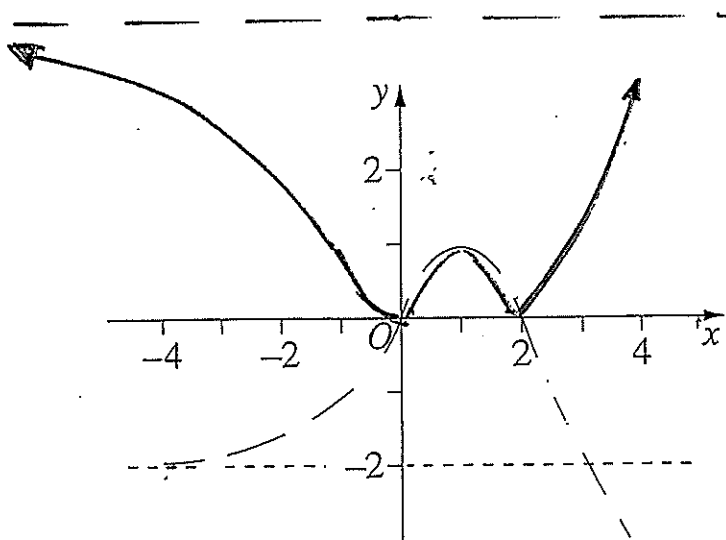
Question 8(a) TEMPLATE SHEET

(i)



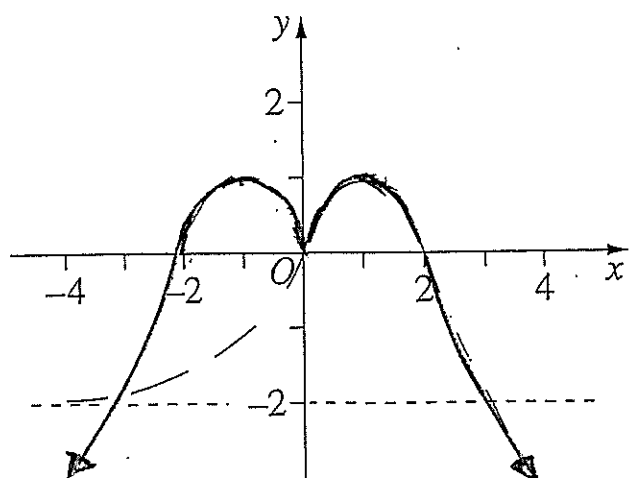
1

(ii)



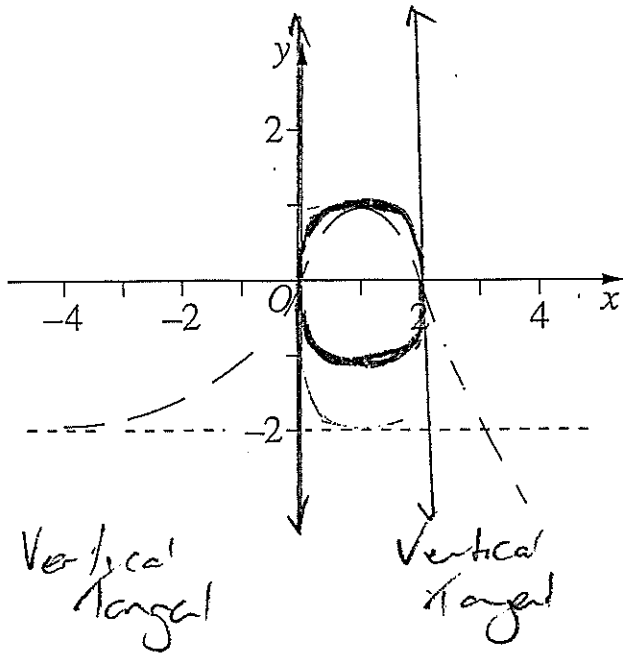
2

(iii)



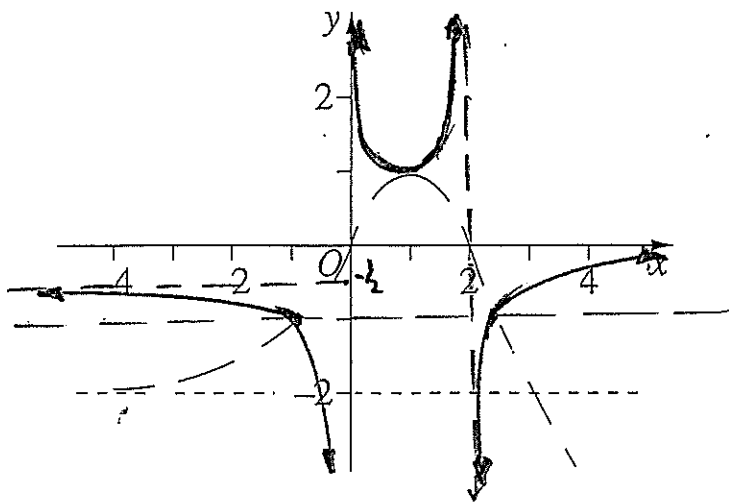
1

(iv)



2

(v)



3